Let (W, S) be a Coxeter group, with $S = \{s_1, \ldots, s_n\}$. I am only going to discuss the case that W is finite, even though significant parts of what I'm going to say can be adapted to the case that W is infinite.

Let $T = \{wsw^{-1}\}$ be the set of reflections of W. Let V be a real n-dimensional vector space, and consider an action of W on V where $t \in T$ acts by the reflection in the hyperplane H_t (which are all distinct).

Let $\mathcal{H} = \{H_t \in T\}$. The convex geometry of this hyperplane arrangement is intimately connected to W, and to a certain order on W called right weak order: $u \lessdot v$ if $v = us$, $s \in S$, and $\ell(v) = \ell(u) + 1$.

$W, <$ is a lattice, that is to say, given two elements u, v, there is a unique smallest element greater than both u and v, called the join of u and v, written $u \vee v$ and a unique largest element smaller than both u and v, called the meet of u and v, written $u \wedge v$. (We will prove this shortly.) A key structural ingredient of a lattice are the join-irreducible elements (those which can't be written non-trivially as a join.) In a finite lattice, being join-irreducible is equivalent to covering exactly one element. If j is join-irreducible, we write j_* for the unique element it covers.

(There is also a dual notion of meet-irreducible element. There is no reason to prefer join-irreducibles to meet-irreducibles, but for my purposes here it suffices to focus on one or the other.)

1. Shards. Our goal in this minicourse will be to study Nathan Reading's *shards*, which are geometrical objects, coming from the reflection arrangement $\mathcal{H}$, which are in natural bijection with the join-irreducible elements of $(W, <)$.

2. Shards and lattice quotients. If L is a lattice, and $\equiv_\Theta$ is an equivalence relation on L which respects the lattice operations, then the equivalence classes L/Θ also have a natural lattice structure. Such a lattice is called a *lattice quotient* of L. One of the reasons to be interested in join-irreducible elements of a lattice is to control lattice quotients.

Specifically, we say that a lattice quotient Θ contracts a join-irreducible j if $j \equiv_\Theta j_*$. A lattice quotient is determined by the set of join-irreducibles it contracts.

Shards help us understand the lattice quotients $(W, <)$: Nathan Reading tells us that the quotient amounts to erasing the shards which are contracted and looking at the induced poset on the resulting regions.

3. Shard polytopes (type A_n). We say that the 1-skeleton of a polytope P in a vector space V realizes a lattice L if there is a bijection $p : x \to p_x$ from L to the vertices of P, such that p_x and p_y are connected by an edge if and only if there is a cover relation between x and y, and there is ρ in V^* such that $x \lessdot y$ iff $\rho(p_y - p_x) > 0$.

There is a beautiful polytope which realizes the right weak order of W: namely, take any point $p \in V$ which lies on no hyperplane, and take the convex hull of its W-orbit. Nathan Reading asked whether any lattice quotient of weak order also has some polytope whose 1-skeleton realizes it. This question was answered in the affirmative by Pilaud and Santos [PS], and again by Padrol, Pilaud, and Ritter [PPR].

Padrol, Pilaud, and Ritter associated certain *shard polytopes* to the shards in type A_n and B_n. I will expalin this in type A_n (from a slightly different perspective). Shard polytopes give a way to answer this question of Reading's: the necessary polytope is the Minkowski sum of the non-contracted shards.

4. Shards and representation theory. Padrol, Pilaud, and Ritter needed to be clever in order to construct the shard polytopes. I will explain how it would have been possible to construct the shard polytopes with no cleverness, using representation theory. In fact, this approach would also have allowed us to discover the definition of shards! What we need is the notion of semistability of representations of a quiver. I am going to explain this first in type A_n, where we can make our lives slightly simpler, at the expense of choosing the algebra that we are using in a way that does not directly extend to other types. I will also explain how one can define shard polytopes in a uniform way in arbitrary simply-laced type (ADE). It remains a conjecture that they have all the good properties of the type A_n shard polytopes.

1. Shards

1.1. **Order on chambers.** In setting out to define shards, it costs nothing to work in a slightly more general setting. Let $\mathcal{H}$ be a finite collection of linear hyperplanes in an n-dimensional real vector space V. The *chambers* of $\mathcal{H}$ are the connected components of $V \setminus \cup_{H\in\mathcal{H}} H$. It is convenient to assume that the collection is reduced, i.e., that $\cap_{H\in\mathcal{H}} H = \{0\}$.

Example 1. *Because we have assumed that our hyperplanes are linear (rather than affine), the pictures we can draw in rank 2 are not very exciting.*

There are two other things we can do to draw higher-dimensional pictures. One is to take an affine slice. This allows us to draw rank three pictures, but it has the disadvantage that we don't see all the chambers.

A third thing we can do is to intersect the hyperplanes with a sphere around the origin, and then stereographically project. The hyperplanes turn into circles.

We choose a chamber B. For any chamber, C, define $s_B(C)$ to the the set of hyperplanes having B and C on opposite sides. We follow [E, BEZ] in ordering the chambers of $\mathcal{H}$ by the inclusion order on $s_B(C)$. We write $L_{\mathcal{H}}$ for the resulting... poset.

Lemma 1. *Cover relations in $L_{\mathcal{H}}$ correspond to pairs of chambers sharing a codimension 1 boundary.*

Proof. It's clear that crossing a hyperplane from one chamber to an adjacent one increases the separation set by one hyperplane, so it is certainly a cover relation. Conversely, if we have two chambers $X > Y$, draw a line from a point in the interior of X to a point in the interior of Y, generically enough that it doesn't pass through any codimension 2 or higher intersections of hyperplanes. Since X and Y are on opposite sides of exactly the hyperplanes in $s_B(X) \setminus s_B(Y)$, the line must cross each of them once, and no others. This gives us an order in which we can pass from X to Y removing one element from the separating set at each step, and thus crossing from one chamber to another via a facet. □

A chamber of $\mathcal{H}$ is called simplicial if it is cut out by n hyperplanes (rather than more). An arrangement is simplicial if all its chambers are simplicial. Any hyperplane arrangement in two-dimensions is simplicial, but this is not true in dimension 3 and higher.

Proposition 1. *If we assume that all the chambers are simplicial, then $L_{\mathcal{H}}$ is a lattice.*

(In fact, this is a stronger assumption than necessary, but it is good enough for our purposes, since reflection arrangements are simplicial.)

To prove this, we will use a result which goes back to [BEZ], and which Reading refers to as the BEZ Lemma.

Lemma 2. *Let L be a finite poset with a minimum element $\hat{0}$ such that whenever x is covered by y and z, then the join $y \vee z$ exists. The L is a lattice.*

This lemma has a fairly simple proof by induction (which I skipped in the lecture).

Proof. The proof is by induction on the size of L. We will first show that for any pair of elements x, y, the join exists. We can assume that x and y are incomparable, since otherwise the join in the larger of them. Let a be an element below x covering $\hat{0}$, and let b be an element below y covering zero. If $a = b$, then we can reduce to the subset of L above a, and we are done by induction. Otherwise, a, b both cover zero, so $a \vee b$ exists by hypothesis. There is a well-defined join of x and $a \vee b$, since both are contained in the part of L above a, and similarly there is a well-defined join of $a \vee b$ and y. Now the join of $x \vee (a \vee b)$ and $(a \vee b) \vee y$ is well-defined, since this is within the part of L above $a \vee b$. Further, this join must be $x \vee y$.

This guarantees that we have joins. To show that $x \wedge y$ exists, take the join of all the elements below x and below y. This set is non-empty, since 0 is in it. The join of all these elements must itself be below x and below y, so it is the meet of x and y. $\square$

Proof of Proposition 1. To apply the BEZ Lemma, we must consider what happens when a chamber x is covered by two other chambers y, z. Necessarily, the picture looks like the rank 2 picutre: the two codimension 1 faces separating x from y and x from z meet in codimension 2 (by the assumption of simplicialness). Clearly the top element in the rank 2 subposet is the join. $\square$

1.2. **Join-irreducible elements of $L_{\mathcal{H}}$.** Recall that a non-minimal element is join-irreducible if it cannot be written as the join of two smaller elements. It is easy to see that this is equivalent to covering exactly one element.

There is a map from join-irreducible chambers to hyperplanes, sending a join-irreducible to the unique hyperplane which one can cross and go down.

Looking again at the rank 2 situation, we see that we can't simply assign a join-irreducible to each hyperplane (as we might have guessed): sometimes two join-irreducibles correspond to the same hyperplane. So, we do the simplest thing imaginable: we split in two the ones that correspond to two join-irreducibles.

In general, what we are going to do is to assign to each join-irreducible which corresponds to a hyperplane H a "shard" of H. These shards will cover H and will be disjoint except on boundaries.

The way this works is as follows. Given several hyperplanes which all intersect in codimension 2, we know what the picture looks like: it is exactly like the two-dimensional case, with a further $n-2$ dimensions in which nothing is happening. Given such a codimension 2 intersection, we temporarily forget all the hyperplanes that do not contain the intersection. The base region B lies in some region of this arrangement with fewer hyperplanes, so we can identify two *basic* hyperplanes, which are the two between which B lies.

Given a hyperplane $H \in \mathcal{H}$, we remove from it those codimension 2 intresections for which H is not basic. Since these intersections are codimension 2 in V, they are codimension 2 in H, so they disconnect H into multiple components.

The closures of these components are the shards.

Example 2. *The A_3 example.*

1.3. **Shards for reflection arrangements.** If we identify the chambers with elements of W by $w \to wB$, then the order on chambers turns onto the weak order.

For each hyperplane, we choose a normal vector in V^* which pairs positively with the points in the base region B. The collection of all these normal vectors, we call Φ^+.

Let $e_1, \dots, e_n$ be the normal vectors corresponding to the hyperplanes adjacent to B. (In the language of root systems, these are the simple roots.)

Hyperplanes that intersection in codimension 2 correspond to normal vectors lying in a 2-plane. The hyperplane H_γ will be split by H_α and H_β if γ is a non-negative linear combination of α and β.

1.3.1. *Shards in type A_n.* Let $\Phi^+ = \{e_i + \dots + e_j \mid i \leq j\}$. This defines the type A_n arrangement. (This convention makes it slightly less obvious that it is really type A_n, giving us the symmetric group $\mathfrak{S}_{n+1}$. On the other hand, this is the uniform perspective.)

Consider the hyperplane $H_{e_i+\dots+e_j}$ for $i \leq j$. A pair of basic hyperplanes cutting this hyperplane are necessarily of the form $H_{e_i+\dots+e_k}, H_{e_{k+1}+\dots+e_j}$. So the hyperplane $e_i + \dots + e_j$ is cut $j - i$ times. Recall that the cuts are codimension 1 inside $H_{e_i+\dots+e_j}$, and it is easy to see that their normals are linearly independent. Thus, $H_{e_i+\dots+e_j}$ is divided into 2^{j-i} shards.

We see that this agrees with what we saw in A_3.

These should correspond to join-irreducible elemens of $\mathfrak{S}_{n+1}$. A join-irreducible corresponding to the hyperplane $e_i + \dots + e_j$ is a permutation which consists of two increasing sets of numbers, the first ending with $j + 1$, the second beginning with i. The elements less than i are necessarily on the left, and those above j are necessarily on the right, so we must decide what to do with the remaining numbers. There are 2^{j-i} choices to make. For future reference, let $w_{(i,j,L,R)}$ be the join-irreducible where L consists of the elements strictly between i and $j+1$ which are in the lefthand sequence, and R consists of those in the righthand sequence

1.3.2. *Shards in type D_4.* Number the vertices of the D_4 diagram 0,1,2,3, so that 0 is the central vertex. For all the roots except the highest one, the situation is as in type A_n: there are at most three cuts, and their normals are linearly independent.

The highest positive root is $2e_0+e_1+e_2+e_3$. The hyperplane $H_{2e_0+e_1+e_2+e_3}$ is cut by the hyperplanes $H_{e_0+e_1}, H_{e_0+e_2+e_3}$, and cyclically — and also the hyperplanes $H_{e_0}, H_{e_0+e_1+e_2+e_3}$. As in type A, the first three pairs of hyperplanes are linearly independant and cut the hyperplane into 8 regions — but then there is an additional hyperplane. It cuts six of the remaining 8 regions, making a total of 14 shards.

1.4. **Shards are in bijection with join-irreducibles.** Let S be a shard of some hyperplane H, and consider the chambers within S. We put an order on them by considering the corresponding $\mathcal{H}$-chambers. The goal will be to show that a minimal chamber in S corresponds to a join-irreducible $\mathcal{H}$-chamber, and that there is exactly one minimal element with respect to this order, thus giving us what we want.

The approach I am taking here is closer to that of [M] than the approach of Reading. Reading's approach uses some more powerful lattice theory, which is lovely, but I would either have had to not prove a bunch of things, or else develop a lot of machinery.

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